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LABSHEET-1

1. Write a Java program to display your name, age, and college name using variables of appropriate data types.

```
public class StudentDetails {
  public static void main(String[] args) {

    String name = "Anubhav Saini
    int age = 18;
    String collegeName = "ABC College";

    System.out.println("Name: " + name);
    System.out.println("Age: " + age);
    System.out.println("College Name: " + collegeName);
  }
}
```

OUTPUT:-

Name: Anubhav Saini

Age: 18

College Name: ABC College

2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).

```
import java.util.Scanner;

public class ArithmeticOperations {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first integer: ");
    int a = sc.nextInt();

    System.out.print("Enter second integer: ");
    int b = sc.nextInt();
```

```
        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));

        sc.close();
    }
}
```

OUTPUT:-

Enter first integer: 20

Enter second integer: 6

Addition: 26

Subtraction: 14

Multiplication: 120

Division: 3

Modulus: 2

3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else.

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 2 == 0) {
            System.out.println(num + " is Even");
        } else {
```

```
            System.out.println(num + " is Odd");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter a number: 15

15 is Odd

4. Write a program to demonstrate the use of relational operators by comparing two numbers.

```
import java.util.Scanner;

public class RelationalOperators {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("a == b: " + (a == b));
        System.out.println("a != b: " + (a != b));
        System.out.println("a > b: " + (a > b));
        System.out.println("a < b: " + (a < b));
        System.out.println("a >= b: " + (a >= b));
        System.out.println("a <= b: " + (a <= b));

        sc.close();
    }
}
```

OUTPUT:-

Enter first number: 10
Enter second number: 5

a == b: false
a != b: true
a > b: true
a < b: false
a >= b: true
a <= b: false

5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.

```
import java.util.Scanner;

public class TypeCasting {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a floating-point number: ");
    double num = sc.nextDouble();

    int converted = (int) num;   // Explicit type casting

    System.out.println("Original value: " + num);
    System.out.println("Converted integer value: " + converted);

    sc.close();
  }
}
```

OUTPUT:-

Enter a floating-point number: 25.75

Original value: 25.75

Converted integer value: 25

6. Write a program to accept a character and print its ASCII value.

```
import java.util.Scanner;

public class ASCIIValue {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a character: ");
    char ch = sc.next().charAt(0);

    int ascii = (int) ch;

    System.out.println("Character: " + ch);
    System.out.println("ASCII Value: " + ascii);

    sc.close();
  }
}
```

OUTPUT:-

Enter a character: A

Character: A

ASCII Value: 65

7. Input a number and check if it is divisible by both 3 and 5 using logical operators.

```
import java.util.Scanner;

public class DivisibleBy3And5 {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");
    int num = sc.nextInt();

    if (num % 3 == 0 && num % 5 == 0) {
```

```
            System.out.println(num + " is divisible by both 3 and 5.");
        } else {
            System.out.println(num + " is not divisible by both 3 and 5.");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter a number: 30

30 is divisible by both 3 and 5.

8. Input a character and check whether it is a vowel or a consonant using if-else.

```
import java.util.Scanner;

public class VowelConsonant {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {

            System.out.println(ch + " is a Vowel.");
        } else {
            System.out.println(ch + " is a Consonant.");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter a character: E

E is a Vowel.

9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria: >=40% in all subjects).

```
import java.util.Scanner;

public class MarksResult {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter marks of Subject 1: ");
    double sub1 = sc.nextDouble();

    System.out.print("Enter marks of Subject 2: ");
    double sub2 = sc.nextDouble();

    System.out.print("Enter marks of Subject 3: ");
    double sub3 = sc.nextDouble();

    double total = sub1 + sub2 + sub3;
    double percentage = total / 3;

    System.out.println("Total Marks: " + total);
    System.out.println("Percentage: " + percentage + "%");
```

```
    if (sub1 >= 40 && sub2 >= 40 && sub3 >= 40) {
      System.out.println("Result: PASS");
    } else {
      System.out.println("Result: FAIL");
    }

    sc.close();
  }
}
```

OUTPUT:-

```
Enter marks of Subject 1: 75
Enter marks of Subject 2: 65
Enter marks of Subject 3: 80

Total Marks: 220.0
Percentage: 73.33333333333333%
Result: PASS
```

10. Write a program to find the greatest of two numbers using if-else.

```
import java.util.Scanner;

public class GreatestOfTwo {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first number: ");
    int a = sc.nextInt();
```

```
        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        if (a > b) {
          System.out.println(a + " is the greatest.");
        } else if (b > a) {
          System.out.println(b + " is the greatest.");
        } else {
          System.out.println("Both numbers are equal.");
        }

        sc.close();
      }
    }
```

OUTPUT:-

Enter first number: 25

Enter second number: 18

25 is the greatest.

11. Input three numbers and print the largest using nested if-else.

```
    import java.util.Scanner;

    public class LargestOfThree {
      public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
```

```
        int c = sc.nextInt();

        if (a > b) {
          if (a > c) {
            System.out.println(a + " is the largest.");
          } else {
            System.out.println(c + " is the largest.");
          }
        } else {
          if (b > c) {
            System.out.println(b + " is the largest.");
          } else {
            System.out.println(c + " is the largest.");
          }
        }

        sc.close();
      }
    }
```

OUTPUT:-

Enter first number: 25

Enter second number: 40

Enter third number: 30

40 is the largest.

12. Accept a year and check whether it is a leap year using if-else.

```
    import java.util.Scanner;

    public class LeapYear {
      public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a year: ");
        int year = sc.nextInt();
```

```
        if (year % 400 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else if (year % 100 == 0) {
            System.out.println(year + " is not a Leap Year.");
        } else if (year % 4 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is not a Leap Year.");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter a year: 2024

2024 is a Leap Year.

13. Accept a number and check whether it is positive, negative, or zero using if-else.

```
import java.util.Scanner;

public class CheckNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num > 0) {
            System.out.println("Positive number");
        } else if (num < 0) {
            System.out.println("Negative number");
        } else {
            System.out.println("Zero");
        }
    }
```

```
}
```

OUTPUT:-

Enter a number: 10

Positive number

14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else

```
import java.util.Scanner;

public class CharacterClassification {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a character: ");
    char ch = sc.next().charAt(0);

    if (ch >= '0' && ch <= '9') {
      System.out.println("It is a digit.");
    } else if (ch >= 'A' && ch <= 'Z') {
      System.out.println("It is an uppercase letter.");
    } else if (ch >= 'a' && ch <= 'z') {
      System.out.println("It is a lowercase letter.");
    } else {
      System.out.println("It is a special character.");
    }
  }
}
```

OUTPUT:-

Enter a character: G

It is an uppercase letter.

15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.

```
import java.util.Scanner;
```

```
public class ArmstrongNumber {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a 3-digit number: ");
    int num = sc.nextInt();

    int original = num;
    int sum = 0;

    int digit1 = num % 10;
    sum = sum + digit1 * digit1 * digit1;
    num = num / 10;

    int digit2 = num % 10;
    sum = sum + digit2 * digit2 * digit2;
    num = num / 10;

    int digit3 = num % 10;
    sum = sum + digit3 * digit3 * digit3;

    if (sum == original) {
      System.out.println(original + " is an Armstrong number.");
    } else {
      System.out.println(original + " is not an Armstrong number.");
    }
  }
}
```

OUTPUT:-

Enter a 3-digit number: 153

153 is an Armstrong number.

16. Input salary and calculate tax based on income slabs using if-else: • Income ≤ 250000: No tax • 250001 to 500000: 5% • 500001 to 1000000: 20% • Above 1000000: 30%

```
import java.util.Scanner;
```

```
public class IncomeTax {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter your salary: ");
      double salary = sc.nextDouble();

      double tax;

      if (salary <= 250000) {
         tax = 0;
      } else if (salary <= 500000) {
         tax = salary * 0.05;
      } else if (salary <= 1000000) {
         tax = salary * 0.20;
      } else {
         tax = salary * 0.30;
      }

      System.out.println("Tax = ₹" + tax);
   }
}
```

OUTPUT:-

```
Enter your salary: 600000
Tax = ₹120000.0
```

17.Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.

```
import java.util.Scanner;


public class PalindromeNumber {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter an integer: ");
      int num = sc.nextInt();


      int original = num;
      int reverse = 0;


      while (num != 0) {
         int digit = num % 10;
         reverse = reverse * 10 + digit;
         num = num / 10;
      }


      if (original == reverse) {
         System.out.println(original + " is a palindrome.");
      } else {
         System.out.println(original + " is not a palindrome.");
      }
   }
```

```
    }
```

OUTPUT:-

```
Enter an integer: 121
121 is a palindrome.
```

18. Accept a number and print whether it is prime or not using if-else.

```
import java.util.Scanner;

public class PrimeNumber {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      boolean isPrime = true;

      if (num <= 1) {
         isPrime = false;
      } else {
         for (int i = 2; i <= num / 2; i++) {
            if (num % i == 0) {
               isPrime = false;
               break;
            }
         }
      }
```

```
        if (isPrime) {
            System.out.println(num + " is a prime number.");
        } else {
            System.out.println(num + " is not a prime number.");
        }
    }
}
```

OUTPUT:-

Enter a number: 17

17 is a prime number.

19. Accept a character and use if-else to identify whether it's a valid operator (+, -, *, /) and perform the corresponding operation on two input numbers.

```
import java.util.Scanner;

public class OperatorCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter an operator (+, -, *, /): ");
```

```
        char operator = sc.next().charAt(0);

        double result;

        if (operator == '+') {
            result = num1 + num2;
            System.out.println("Result = " + result);
        } else if (operator == '-') {
            result = num1 - num2;
            System.out.println("Result = " + result);
        } else if (operator == '*') {
            result = num1 * num2;
            System.out.println("Result = " + result);
        } else if (operator == '/') {
            if (num2 != 0) {
                result = num1 / num2;
                System.out.println("Result = " + result);
            } else {
                System.out.println("Division by zero is not allowed.");
            }
        } else {
            System.out.println("Invalid operator.");
        }
    }
}
```

OUTPUT:-

Enter first number: 20

Enter second number: 5

Enter an operator (+, -, *, /): *

Result = 100.0

20. Accept age and gender and determine eligibility for marriage (male ≥ 21, female ≥ 18) using nested if-else conditions.

```
import java.util.Scanner;


public class MarriageEligibility {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter your age: ");
      int age = sc.nextInt();


      System.out.print("Enter your gender (M/F): ");
      char gender = sc.next().charAt(0);


      if (gender == 'M' || gender == 'm') {
         if (age >= 21) {
            System.out.println("Eligible for marriage.");
         } else {
            System.out.println("Not eligible for marriage.");
         }
      } else if (gender == 'F' || gender == 'f') {
         if (age >= 18) {
```

```
            System.out.println("Eligible for marriage.");
        } else {
            System.out.println("Not eligible for marriage.");
        }
    } else {
        System.out.println("Invalid gender.");
    }
  }
}
```

OUTPUT:-

Enter your age: 22

Enter your gender (M/F): M

Eligible for marriage.

21. Accept three angles and check whether they can form a valid triangle (sum of angles = 180°).

```
import java.util.Scanner;

public class ValidTriangle {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first angle: ");
    int angle1 = sc.nextInt();

    System.out.print("Enter second angle: ");
    int angle2 = sc.nextInt();
```

```
        System.out.print("Enter third angle: ");
        int angle3 = sc.nextInt();

        if (angle1 > 0 && angle2 > 0 && angle3 > 0 &&
            angle1 + angle2 + angle3 == 180) {
            System.out.println("The angles can form a valid triangle.");
        } else {
            System.out.println("The angles cannot form a valid triangle.");
        }
    }
}
```

OUTPUT:-

```
Enter first angle: 60
Enter second angle: 60
Enter third angle: 60
The angles can form a valid triangle.
```

22. Input the length and breadth of a rectangle, and determine whether it is a square using if-else.

```
import java.util.Scanner;

public class SquareCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter length: ");
```

```
        int length = sc.nextInt();

        System.out.print("Enter breadth: ");
        int breadth = sc.nextInt();

        if (length == breadth) {
            System.out.println("The rectangle is a square.");
        } else {
            System.out.println("The rectangle is not a square.");
        }
    }
}
```

OUTPUT:-

```
Enter length: 10
Enter breadth: 10
The rectangle is a square.
```

23. Input a temperature in Celsius and convert it to Fahrenheit using type casting and expressions.

```
import java.util.Scanner;

public class CelsiusToFahrenheit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();
```

```
        double fahrenheit = (celsius * 9 / 5) + 32;

        System.out.println("Temperature in Fahrenheit = " + fahrenheit);
    }
}
```

OUTPUT:-

Enter temperature in Celsius: 25

Temperature in Fahrenheit = 77.0

24. Accept a number and check whether it is a perfect number (sum of divisors = number) using basic arithmetic and if-else.

```
import java.util.Scanner;

public class PerfectNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int sum = 0;

        for (int i = 1; i < num; i++) {
            if (num % i == 0) {
                sum = sum + i;
            }
```

```
        }

        if (sum == num) {
            System.out.println(num + " is a perfect number.");
        } else {
            System.out.println(num + " is not a perfect number.");
        }
    }
}
```

OUTPUT:-

Enter a number: 28

28 is a perfect number.

25. Accept two integers and a logical operator as a character. Use if-else to perform the logical operation and display the result.

```
import java.util.Scanner;

public class LogicalOperation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second integer: ");
        int num2 = sc.nextInt();
```

```
        System.out.print("Enter logical operator (&, |, ^): ");
        char operator = sc.next().charAt(0);

        int result;

        if (operator == '&') {
            result = num1 & num2;
            System.out.println("Result = " + result);
        } else if (operator == '|') {
            result = num1 | num2;
            System.out.println("Result = " + result);
        } else if (operator == '^') {
            result = num1 ^ num2;
            System.out.println("Result = " + result);
        } else {
            System.out.println("Invalid logical operator.");
        }
    }
}
```

OUTPUT:-

Enter first integer: 5

Enter second integer: 3

Enter logical operator (&, |, ^): &

Result = 1

26. Input a five-digit number and check whether the first and last digits are the same using arithmetic expressions.

```
import java.util.Scanner;


public class FirstLastDigit {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter a five-digit number: ");
      int num = sc.nextInt();


      int firstDigit = num / 10000;
      int lastDigit = num % 10;


      if (firstDigit == lastDigit) {
         System.out.println("First and last digits are the same.");
      } else {
         System.out.println("First and last digits are not the same.");
      }
   }
}
```

OUTPUT:-

```
Enter a five-digit number: 12321
First and last digits are the same.
```

27. Input the cost price and selling price of an item. Determine profit or loss, and by how much, using if-else.

```
import java.util.Scanner;
```

```
public class ProfitLoss {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter cost price: ");
      double costPrice = sc.nextDouble();

      System.out.print("Enter selling price: ");
      double sellingPrice = sc.nextDouble();

      if (sellingPrice > costPrice) {
         double profit = sellingPrice - costPrice;
         System.out.println("Profit = " + profit);
      } else if (costPrice > sellingPrice) {
         double loss = costPrice - sellingPrice;
         System.out.println("Loss = " + loss);
      } else {
         System.out.println("No profit, no loss.");
      }
   }
}
```

OUTPUT:-

```
Enter cost price: 500
Enter selling price: 650
Profit = 150.0
```

28. Accept a number and display the number of digits using expressions and if-else.

```
import java.util.Scanner;

public class NumberOfDigits {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      int n = Math.abs(num);

      if (n < 10) {
         System.out.println("Number of digits = 1");
      } else if (n < 100) {
         System.out.println("Number of digits = 2");
      } else if (n < 1000) {
         System.out.println("Number of digits = 3");
      } else if (n < 10000) {
         System.out.println("Number of digits = 4");
      } else {
         System.out.println("Number of digits = 5 or more");
      }
   }
}
```

OUTPUT:-

```
Enter a number: 4567
```

Number of digits = 4

29. Input the height (in cm) of a person and classify them as short (170).

```
import java.util.Scanner;


public class HeightClassification {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter height in cm: ");
      int height = sc.nextInt();


      if (height < 150) {
         System.out.println("Short");
      } else if (height <= 170) {
         System.out.println("Average");
      } else {
         System.out.println("Tall");
      }
   }
}
```

OUTPUT:-

Enter height in cm: 165

Average

30. Write a Java program to simulate a simple calculator that takes two numbers and an operator (+, , *, /) and performs the corresponding operation using if-else.

```
import java.util.Scanner;
```

```
public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (num1 + num2));
        } else if (operator == '-') {
            System.out.println("Result = " + (num1 - num2));
        } else if (operator == '*') {
            System.out.println("Result = " + (num1 * num2));
        } else if (operator == '/') {
            if (num2 != 0) {
                System.out.println("Result = " + (num1 / num2));
            } else {
                System.out.println("Cannot divide by zero.");
            }
```

```
        } else {
            System.out.println("Invalid operator.");
        }
    }
}
```

OUTPUT:-

```
Enter first number: 20
Enter second number: 5
Enter operator (+, -, *, /): /
Result = 4.0
```